

# Lakshya JEE 2.0 2025

## Maths

DPP: 3

### Method of Differentiation

**Q1** The derivative of  $e^{x^3}$  with respect to  $\log x$  is

- (A)  $e^{x^3}$   
 (B)  $3x^2 2e^{x^3}$   
 (C)  $3x^3 e^{x^3}$   
 (D)  $3x^2 e^{x^3} + 3x^2$

**Q2** The rate of change of  $\sqrt{(x^2 + 16)}$  with respect to  $\frac{x}{x-1}$  at  $x = 3$  is

- (A) 2  
 (B)  $\frac{11}{5}$   
 (C)  $-\frac{12}{5}$   
 (D) -3

**Q3** If  $x = \log(1 + t^2)$  and  $y = t - \tan^{-1} t$ . Then,  $\frac{dy}{dx}$  is equal to

- (A)  $e^x - 1$   
 (B)  $t^2 - 1$   
 (C)  $\frac{\sqrt{e^x - 1}}{2}$   
 (D)  $e^x - y$

**Q4** Differentiate  $f(\sin x)$  w.r.t.  $f(\cos x)$

- (A)  $\frac{f'(\sin x) \cos x}{f'(\cos x) \sin x}$   
 (B)  $\frac{f'(\cos x) \sin x}{f'(\sin x) \cos x}$   
 (C)  $-\frac{f'(\cos x) \sin x}{f'(\sin x) \cos x}$   
 (D)  $\frac{f'(\cos x) \sin x}{f'(-\sin x) \cos x}$

**Q5** If  $\sin y + e^{-x \cos y} = e$ , then  $\frac{dy}{dx}$  at  $(1, \pi)$  is

- (A)  $\sin y$   
 (B)  $-x \cos y$   
 (C)  $e$   
 (D)  $\sin y - x \cos y$

**Q6** If  $y = \cos^2 \frac{3x}{2} - \sin^2 \frac{3x}{2}$ , then  $\frac{d^2 y}{dx^2}$  is

- (A)  $-3\sqrt{1 - y^2}$   
 (B)  $9y$   
 (C)  $-9y$   
 (D)  $3\sqrt{1 - y^2}$

**Q7** If  $y = A \cos(\log x) + B \sin(\log x)$ , then

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y =$$

- (A) 1 (B) 0  
 (C) -1 (D) 2

**Q8** If

$$f(x) = 1 + nx + \frac{n(n-1)}{2}x^2 + \frac{n(n-1)(n-2)}{6}x^3 + \dots +$$

$x^n$ , then  $f''(1)$  is equal to

- (A)  $n(n-1)2^{n-1}$   
 (B)  $(n-1)2^{n-1}$   
 (C)  $n(n-1)2^{n-2}$   
 (D)  $n(n-1)2^n$

**Q9**  $\frac{d}{dx} \left[ \tan^{-1} \left( \frac{a-x}{1+ax} \right) \right]$  is equal to

- (A)  $-\frac{1}{1+x^2}$   
 (B)  $\frac{1}{1+a^2} - \frac{1}{1+x^2}$   
 (C)  $\frac{1}{1+\left(\frac{a-x}{1+ax}\right)^2}$   
 (D)  $\frac{-1}{\sqrt{1-\left(\frac{a-x}{1+ax}\right)^2}}$

**Q10** If  $y = \tan^{-1} \left( \frac{a \cos x - b \sin x}{b \cos x + a \sin x} \right)$ , then  $\frac{dy}{dx}$  is equal to

- (A) 2  
 (B) -1



- (C)  $\frac{a}{b}$   
 (D)  $\frac{b}{a}$

**Q11** If  $y = \sec^{-1}\left(\frac{\sqrt{x}+1}{\sqrt{x}-1}\right) + \sin^{-1}\left(\frac{\sqrt{x}-1}{\sqrt{x}+1}\right)$ , then

$\frac{dy}{dx}$  equals.

- (A) 1  
 (B) 0  
 (C)  $\frac{\sqrt{x}+1}{\sqrt{x}-1}$   
 (D)  $\frac{\sqrt{x}-1}{\sqrt{x}+1}$

**Q12** If  $x = 2 \sin \theta - \sin 2\theta$  and

$y = 2 \cos \theta - \cos 2\theta$ ,  $\theta \in [0, 2\pi]$ , then  $\frac{d^2y}{dx^2}$  at

$\theta = \pi$  is:

- (A)  $\frac{3}{8}$   
 (B)  $\frac{3}{2}$   
 (C)  $\frac{3}{4}$   
 (D)  $-\frac{3}{4}$

**Q13** The derivative of  $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$  with respect to  $\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)$  at  $x = \frac{1}{2}$  is:

- (A)  $\frac{2\sqrt{3}}{3}$   
 (B)  $\frac{2\sqrt{3}}{5}$   
 (C)  $\frac{\sqrt{3}}{10}$   
 (D)  $\frac{\sqrt{3}}{12}$

**Q14** Let  $y = y(x)$  be a function of  $x$  satisfying  $y\sqrt{1-x^2} = k - x\sqrt{1-y^2}$  where  $k$  is constant and  $y\left(\frac{1}{2}\right) = -\frac{1}{4}$ . Then  $\frac{dy}{dx}$  at  $x = \frac{1}{2}$ , is equal to:

- (A)  $\frac{5}{\sqrt{7}}$   
 (B)  $-\frac{5}{\sqrt{4}}$   
 (C)  $-\frac{5}{\sqrt{2}}$   
 (D)  $-\frac{\sqrt{5}}{2}$



## Answer Key

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Q1 (C)

Q2 (C)

Q3 (C)

Q4 (A)

Q5 (C)

Q6 (C)

Q7 (B)

Q8 (C)

Q9 (A)

Q10 (B)

Q11 (B)

Q12 (A)

Q13 (C)

Q14 (D)

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